PROBLEM STATEMENT -1 (WEB-3)

Personal Onchain AI Treasurer:-

Problem Description:-

With the increasing adoption of decentralized finance (DeFi), users manage assets across multiple platforms for savings, yield generation, subscriptions, and payments. However, there is no unified intelligent system that can autonomously manage these activities while ensuring transparency, security, and user control. Users often struggle with fragmented financial data, inefficient yield strategies, unmanaged subscriptions, and lack of budgeting insights.

Objective:-

To develop an AI-powered onchain agent that autonomously manages personal crypto finances, including savings, yield optimization, subscriptions, liquidity positions, bill payments, and daily budgeting, while providing explainable and auditable decision-making.

Key Requirements:-

Manage user savings and fund allocation across DeFi protocols

Optimize yield strategies (staking, lending, LP positions)

Track and automate subscriptions and bill payments

Maintain and analyze daily/weekly budgets

Provide AI-driven recommendations with clear explanations

Ensure auditability and transparency of all decisions

Constraints:-

Must be non-custodial

Should work on EVM-compatible blockchain

Ensure security, privacy, and gas efficiency

Expected Outcome:-

A functional prototype of an AI-based onchain treasurer that automates financial management, improves decision-making, and enhances user control with transparent and explainable operations.

PROBLEM STATEMENT - 2

Infrastructure for Autonomous Onchain Agents

Problem Description:-

With the emergence of AI-driven systems and decentralized applications, autonomous agents are increasingly expected to perform onchain actions such as trading, governance participation, and asset management. However, there is a lack of standardized infrastructure to securely deploy, manage, and monitor such agents based on user-defined intents. Existing systems lack reliability, transparency, and control mechanisms, making autonomous execution risky and inefficient.

Objective:-

To build a robust infrastructure that enables deployment and management of autonomous agent “bots” capable of executing onchain actions based on user-defined intents, while ensuring security, transparency, and controllability.

Key Requirements:-

Enable deployment of autonomous onchain agents (bots)

Allow users to define intents (e.g., trade, transfer, rebalance)

Agents should execute actions independently based on conditions

Provide monitoring and control layer (pause, modify, override)

Ensure secure transaction execution and permissioning

Maintain logs and explainability of agent decisions

Constraints:-

Must be compatible with EVM-based blockchains

Ensure non-custodial architecture

Minimize security risks and unauthorized actions

Expected Outcome:-

A scalable infrastructure that allows users to deploy and manage autonomous agents capable of executing reliable, secure, and explainable onchain operations based on predefined intents.

PROBLEM STATEMENT -3

Decentralized Agent Marketplace

Problem Description:-

As AI and blockchain technologies evolve, autonomous agents are being developed for tasks such as trading, farming, scheduling, and business automation. However, there is no unified platform where users can easily discover, deploy, and utilize such agents in a secure and decentralized manner. The absence of a standardized marketplace limits accessibility, interoperability, and scalability of agent-based solutions.

Objective:-

To build a decentralized marketplace that enables users to discover, deploy, and interact with various AI-powered agents for different use cases, while ensuring security, transparency, and ease of use.

Key Requirements:-

Develop a marketplace to host multiple types of agents, including:

Trading agent

Farming (yield) agent

Scheduling agent

Portfolio rebalancing agent

Content reply agent

Business assistant agent

Enable users to select, configure, and deploy agents

Provide rating, reputation, or verification system for agents

Ensure secure and permissioned interactions

Maintain logs and transparency of agent actions

Constraints:-

Must support decentralized architecture (Web3 compatible)

Ensure data privacy and non-custodial operations

Agents should operate within user-defined permissions

Expected Outcome:-

A functional decentralized marketplace where users can access and utilize diverse AI agents efficiently, enabling automation across financial, personal, and business domains.

PROBLEM STATEMENT -4

Agentic Protocol for Auditability and Explainability

Problem Description:-

As autonomous agents increasingly perform critical tasks and make decisions with real-world consequences, there is a lack of standardized mechanisms to ensure transparency, accountability, and trust. Current systems do not provide clear insights into why a decision was made, nor do they allow humans or governing bodies to verify or challenge agent actions. This creates risks in high-stakes environments such as finance, governance, and automation.

Objective:-

To design and develop a protocol that enables transparent logging, explainability, and verification of decisions made by autonomous agents.

Key Requirements:-

Log all agent decisions and actions in a structured manner

Provide human-readable explanations for each decision

Enable verification and auditing of agent behavior

Allow users or DAOs to challenge or override decisions

Ensure tamper-proof and transparent record keeping

Constraints:-

Must support decentralized and trustless environments

Ensure data integrity and security

Should be scalable for multiple agents and high-frequency actions

Expected Outcome:-

A robust protocol that enhances trust in autonomous agents by enabling full transparency, explainability, and governance over their decisions.

PROBLEM STATEMENT -5

AI Wallet Assistant for Transaction Understanding and Risk Awareness

Problem Description:-

New users entering the Web3 ecosystem often find it difficult to understand wallet activities, transaction details, and smart contract interactions. This lack of clarity creates a sense of risk and leads to mistakes such as signing malicious transactions or interacting with unsafe contracts. There is a need for an intelligent system that simplifies wallet interactions and enhances user confidence.

Objective:-

To develop an AI-powered wallet assistant that helps users understand transactions, assess risks, and make informed decisions before interacting with blockchain applications.

Key Requirements:-

Explain wallet transactions in simple, human-readable language

Answer user queries like “What will happen if I sign this?”

Detect suspicious or malicious contract interactions

Offer guidance and recommendations for safer actions

Constraints:-

Must ensure user privacy and data security

Should integrate with existing crypto wallets (EVM compatible)

Provide fast and accurate responses in real-time

Expected Outcome:-

A user-friendly AI assistant that improves transparency, reduces risk, and enhances trust for users interacting with blockchain wallets.

PROBLEM STATEMENT - 6

Intent-Based Token Swap using AI Agents

Problem Description:-

Token swapping in decentralized finance (DeFi) often involves multiple steps, manual comparisons across platforms, and technical complexity, making it inefficient and confusing for users. There is a lack of intuitive systems that allow users to express simple intents while automatically handling route optimization, slippage, and execution securely.

Objective:-

To develop an intent-based system where users can perform token swaps using natural language commands, while an intelligent agent handles route discovery, optimization, and execution.

Key Requirements:-

Allow users to input natural language intents (e.g., “Convert 1 ETH to USDC”)

Identify and select the best swap route across DEXs

Support trade simulation before execution

Optimize for best rate, lowest fees, and minimal slippage

Enable secure transaction execution with user approval

Provide clear explanation of the swap process

Constraints:-

Must support EVM-compatible blockchains

Ensure non-custodial and secure transactions

Maintain real-time performance and accuracy

Expected Outcome:-

A seamless token swapping experience where users interact through simple intents, while the system efficiently executes optimized trades with transparency and minimal effort.

PROBLEM STATEMENT - 7

AI + Web3 Event Matchmaker

Problem Description:-

At large tech and networking events, participants often struggle to connect with the right people who align with their interests, goals, or expertise. Existing networking methods are unstructured and inefficient, leading to missed opportunities for meaningful collaboration. There is a need for an intelligent and decentralized system that facilitates targeted and effective networking.

Objective:-

To develop an AI-powered matchmaking platform that connects event participants based on shared interests, goals, and profiles, enabling more meaningful and productive interactions.

Key Requirements:-

Allow users to create profiles with interests, skills, and goals

Use AI to match participants with high relevance

Suggest people to meet during the event

Optionally generate personalized introduction messages

Ensure privacy and user-controlled data sharing (Web3-based)

Provide a simple interface for real-time interaction and discovery

Constraints:-

Must ensure data privacy and user consent

Should support scalable usage during live events

Prefer integration with Web3 identity or wallet systems

Expected Outcome:-

A smart networking platform that enhances event experiences by enabling users to discover and connect with the most relevant people efficiently.

PROBLEM STATEMENT - 8

AI-Powered Opportunity Discovery Platform (Spotify for Opportunities)

Problem Description:-

Students and professionals often struggle to discover relevant opportunities such as jobs, projects, airdrops, and communities due to scattered platforms and lack of personalization. Existing systems do not effectively adapt to user preferences or leverage on-chain activity for tailored recommendations. There is a need for a unified, intelligent platform that curates opportunities in a personalized and engaging manner.

Objective:-

To develop an AI-powered platform that recommends personalized opportunities to users based on their interests, behavior, and on-chain activity, providing an intuitive and engaging discovery experience.

Key Requirements:-

Provide a feed-based interface to explore opportunities (jobs, projects, airdrops, communities)

Enable swipe-based interaction (like/dislike/save)

Use AI to learn user preferences and improve recommendations

Display suggestions like “Because you explored X, try this”

Integrate wallet/on-chain data for better personalization

Ensure user privacy and control over data usage

Constraints:-

Must ensure secure handling of user and wallet data

Should scale for large datasets and real-time recommendations

Prefer integration with Web3 identity systems

Expected Outcome:-

A smart recommendation platform that delivers highly personalized opportunities, improving discovery, engagement, and career growth through AI and Web3 integration.

PROBLEM STATEMENT - 9

Duolingo for Web3: Gamified Learning with Onchain Proofs

Problem Description:-

Web3 concepts such as cryptocurrencies, DeFi, and NFTs are often difficult for beginners to understand due to lack of practical, guided learning platforms. Existing resources are either too theoretical or lack hands-on experience, resulting in low engagement and poor skill retention. There is a need for an interactive system that combines learning with real onchain actions in a simplified and engaging way.

Objective:-

To develop a gamified learning platform that teaches Web3 concepts through interactive tasks, real-world simulations, and adaptive AI-driven guidance.

Key Requirements:-

Provide structured learning for crypto basics, DeFi, and NFTs

Implement learn → practice → earn model

Include hands-on tasks like:

Token swaps (testnet)

NFT minting

Generate onchain proofs/certificates of completion

Use AI to adapt difficulty and personalize learning paths

Ensure an engaging, gamified user experience

Constraints:-

Must operate on testnets to avoid financial risk

Ensure user-friendly onboarding for beginners

Maintain data privacy and secure wallet interactions

Expected Outcome:-

An interactive Web3 learning platform that enhances understanding through real actions, increases engagement via gamification, and provides verifiable onchain achievements.

PROBLEM STATEMENT - 10

Social Wallet: Onchain Activity Feed and Insights

Problem Description:-

In the Web3 ecosystem, users often lack visibility into meaningful onchain activities within their trusted network, making it difficult to identify trends, opportunities, or informed actions. Existing wallets focus only on individual transactions and do not provide social or contextual insights. There is a need for a platform that brings a social layer to wallets, enabling users to learn and act based on trusted network activity.

Objective:-

To develop a social wallet system that provides a personalized feed of onchain activities from trusted users, helping individuals discover trends, opportunities, and insights within their networK

Key Requirements:-

Display a feed of onchain activities from selected/trusted users

Highlight trends and patterns within the user's network

Allow users to follow, filter, or customize their feed

Provide insights and summaries of activities (e.g., popular tokens, actions)

Ensure privacy controls and selective data sharing

Offer simple explanations of observed activities

Constraints:-

Must ensure user consent and privacy protection

Should support scalable data processing for real-time feeds

Prefer integration with Web3 wallets and identity systems

Expected Outcome:-

A social-enabled wallet experience that enhances decision-making by surfacing relevant onchain activities and trends from trusted networks.